

Problem

The current through a 0.5-F capacitor is $6(1 - e^{-t})$ A. Determine the voltage and power at $t = 2$ s. Assume $v(0) = 0$.

Step-by-step solution

Step 1 of 4

The expression for the voltage across the capacitor is,

$$v(t) = \frac{1}{C} \int i(t) dt \dots\dots (1)$$

Here, $i(t)$ is the current flowing through the capacitor and C is the capacitance.

Substitute $6(1 - e^{-t})$ for $i(t)$, and 0.5 for C in equation (1).

$$\begin{aligned} v(t) &= \frac{1}{0.5} \int 6(1 - e^{-t}) dt \\ &= 12 \int (1 - e^{-t}) dt \end{aligned}$$

$$v(t) = 12(t + e^{-t}) + c \dots\dots (2)$$

At $t = 0$,

$$\begin{aligned} v(t) &= v(0) \\ &= 0 \text{ V} \end{aligned}$$

Step 2 of 4

From equation (2),

$$v(t) = 12(t + e^{-t}) + c$$

Substitute 0 for t .

$$v(0) = 12(0 + e^{-0}) + c$$

$$0 = 12 + c$$

$$c = -12$$

Thus,

$$v(t) = 12(t + e^{-t}) - 12 \dots\dots (3)$$

Step 3 of 4

Substitute 2 for t in equation (3).

$$\begin{aligned} v(2) &= 12(2 + e^{-2}) - 12 \\ &= 12(2 + 0.135) - 12 \\ &= 13.62 \text{ V} \end{aligned}$$

Therefore, the voltage across the capacitor at $t = 2$ s is 13.62 V.

Step 4 of 4

The expression for the current through the capacitor is,

$$i(t) = 6(1 - e^{-t}) \text{ A}$$

Determine the current at $t = 2 \text{ s}$.

$$\begin{aligned} i(2) &= 6(1 - e^{-2}) \\ &= 6(1 - 0.135) \\ &= 5.19 \text{ A} \end{aligned}$$

The expression for the power is,

$$p(t) = i(t)v(t)$$

Determine the power at $t = 2 \text{ s}$.

$$\begin{aligned} p(2) &= i(2)v(2) \\ &= (5.19)(13.62) \\ &= 70.69 \text{ W} \end{aligned}$$

Therefore, the power at $t = 2 \text{ s}$ is 70.69 W.